AgCalculator

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Grain Bin Capacity Worksheet

Printable field reference

Fill this in at the calculator online, or work it by hand with the formulas below. Estimates only — verify measured test weight and bin geometry before relying on results.

1. Your inputs

Grain type	_____	Roof type (circle one):	_____
Test weight (lb/bu)	_____	☐ Flat ☐ Conical	
Moisture shrink (%)	_____	Conical roof height (ft)	_____
Bin diameter (ft)	_____	Hopper height, optional (ft)	_____
Straight-wall height (ft)	_____	Reverse: bushels → fill height?	_____

2. Formulas

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Cylinder volume (ft³) = π × (D/2)² × H
Conical roof / hopper (ft³) = π × (D/2)² × height ÷ 3
Bushels = total ft³ ÷ 1.2445 (ft³ per U.S. bushel)
Weight (lb) = bushels × test weight (lb/bu)  Tons = pounds ÷ 2000
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Moisture shrink reduces calculated bushels by the chosen percent.

3. Worked example

36 ft diameter, 24 ft straight wall, conical roof 4 ft, corn at 56 lb/bu, no shrink:
Cylinder: π×18²×24 ≈ **24,429 ft³** · Roof cone: π×18²×4÷3 ≈ **1,357 ft³**
Total ≈ 25,786 ft³ → Bushels: 25,786÷1.2445 ≈ **20,713 bu**
Weight: 20,713×56 ≈ **1,160,000 lb (~580 tons)**

4. Your numbers

Cylinder volume	π(D/2)²H	_____
Roof/hopper volume	π(D/2)²h÷3	_____
Total volume (ft³)	sum	_____
Bushels	ft³ ÷ 1.2445	_____
Weight (lb)	bushels × test wt	_____
Weight (tons)	lb ÷ 2000	_____

Notes